

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 1.2: Indices — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.2: Indices
Driving Question	DRIVING QUESTION: Why do scientists, economists and farmers use "small superscript numbers" to write huge or tiny quantities — and what rules make those compact symbols so powerful for computation?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Noticing the Problem — Anchoring the Phenomenon of Indices	Two cards side by side: Card A shows Kenya's 2019 Census figure 47,564,296 written in full; Card B shows the approximation 4.8×10^7 . Learners also observed that numbers like 64 (counties originally planned), 729 (calories in ugali and sukuma wiki) and 100,000 (approximate population of Kericho) can each be written as a short chain of repeated prime-factor multiplications.	A number written in index form has two parts: a BASE (the number being repeatedly multiplied) and an INDEX or POWER (the count of how many times the base is multiplied by itself). Index notation is a compact, unambiguous shorthand for repeated multiplication — the same shorthand that journalists, census planners and scientists use when working with Kenya's large national data.	We can explain why Card B is more compact than Card A: the index 7 in 10^7 tells us exactly how many times 10 is multiplied by itself, replacing seven individual multiplication symbols. We cannot yet explain what happens when two index expressions are multiplied or divided, why any number to the power zero equals 1, or how to use index notation for very small numbers like the mass of a maize grain — those questions are now posted on our Driving Question Board.
Lesson 2 Building Index Language: Bases, Exponents and Expanded Form	Learners examined County Population Cards (Nairobi $\approx 4.4 \times 10^6$, Kisumu $\approx 1.2 \times 10^6$, Mombasa $\approx 1.2 \times 10^6$, Turkana $\approx 9.3 \times 10^5$) from the KNBS 2019 Census and a KNBS maize harvest figure ($\approx 3.5 \times 10^6$ tonnes). They also observed six index expression cards (10^3 , 2^5 , 3^4 , 5^3 , 10^6 , 10^7) and expanded each into repeated multiplication strings.	Learners learned that an index expression a^n means the base a multiplied by itself n times (e.g. $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$). They learned to identify the BASE (number being multiplied) and the EXPONENT or INDEX or POWER (count of multiplications), and to write large real-world quantities — county populations and agricultural tonnage — in the compact form $a \times 10^n$ (standard index form).	Learners explained why the 2019 Census figure for Kenya's national population ($\approx 4.8 \times 10^7$) uses a superscript 7: because $10^7 = 10,000,000$ (ten million), and the population is approximately 4.8 ten-millions. They connected the exponent directly to the number of times 10 is multiplied by itself, and articulated a rule linking the exponent to the magnitude of the number — building the conceptual foundation for the product and quotient laws of indices in Lessons 3 and 4.

Lesson 3 Discovering the Product Law of Indices	Learners expanded products of same-base powers (e.g. $2^3 \times 2^4$; $10^2 \times 10^3$) step by step in a structured Expansion Table, counting all factors and recording results in index form across six rows including an abstract base row.	When two powers with the same base are multiplied, the exponent of the result equals the sum of the two exponents: $a^m \times a^n = a^{m+n}$. This was confirmed numerically, algebraically, and through a Kericho tea-farm dilution scenario involving powers of 10.	The Product Law works because multiplying a^m by a^n places m copies of the base and n copies of the base into one continuous product — a total of $m + n$ identical factors — which by definition equals a^{m+n} . This is why multiplying two large numbers written as powers of 10 (such as Kenyan census figures) only requires adding two small exponents rather than performing full multi-digit multiplication.
Lesson 4 Discovering the Quotient Law of Indices	Learners expanded same-base division problems (e.g. $2^5 \div 2^2$, $10^6 \div 10^4$, $3^4 \div 3^1$) in full and cancelled common factors in the numerator and denominator, recording the pattern in a structured four-column Discovery Card table.	Learners deduced the Quotient Law of Indices ($a^m \div a^n = a^{m-n}$) from the cancellation pattern, applied it to a Uasin Gishu maize-harvest scenario, and discovered the Zero-Index Law ($a^0 = 1$) as the edge case when $m = n$ — deriving it from both the Quotient Law and ordinary arithmetic.	Learners formalised the Quotient Law in symbolic form, presented their deductions to peers, added the law and the a^0 wonder to the Driving Question Board, and updated their cumulative Individual Index-Law Models with a Division Arrow and an a^0 node marked with a question mark to signal ongoing inquiry.
Lesson 5 Discovering the Power-of-a-Power Law: $(a^m)^n = a^{mn}$	Learners expanded expressions such as $(2^3)^2$, $(5^3)^2$, and $(2^{10})^3$ by writing out every individual factor, then counted the total number of factors to express the result as a single power. They also read a Safaricom data card stating that peak M-Pesa transactions could be expressed as $(10^3)^2$ and simplified it.	When a power is raised to another power, the two exponents are multiplied: $(a^m)^n = a^{mn}$. This is because writing a^m a total of n times produces $m \times n$ individual base factors. This law means $(2^{10})^3 = 2^{30}$ — so 1 GB $\approx 2^{30}$ bytes — without ever multiplying out the full number. All four core laws of indices (product, quotient, zero-index, power-of-a-power) are now confirmed and arise from the single idea of repeated multiplication.	The power-of-a-power law explains why scientists, economists, and Kenyan government planners can manipulate huge quantities — like census figures, smartphone storage sizes, or M-Pesa transaction volumes — using only small superscript numbers and simple arithmetic on those exponents. It also explains the compact notation on the anchoring phenomenon card: the superscript 7 in 4.8×10^7 is not arbitrary — it belongs to a system of four interlocking laws, all provable from first principles, that make computation with very large (and very small) numbers both exact and efficient.
Lesson 6 Consolidating and Applying All Four Laws of Indices — Multi-Step Computations	Learners observed that multi-step index computations — such as $(3^2 \times 3^5) \div 3^4$ and $((2^3)^2 \times 2^4) \div 2^6$ — require selecting and combining more than one law in a deliberate sequence, and that digital tools on the Kenya Education Cloud confirm the same answers produced by applying the laws by hand. They also observed that Kenya's 2019 census population (4.8×10^7) can be projected forward by multiplying index expressions using the Product Law alone.	The four laws of indices — Product ($a^m \times a^n = a^{m+n}$), Quotient ($a^m \div a^n = a^{m-n}$), Zero-Index ($a^0 = 1$), and Power-of-a-Power ($(a^m)^n = a^{mn}$) — form a coherent toolkit that can be combined in sequence to simplify any multi-step index expression. The Power-of-a-Power Law must be resolved before the Product or Quotient Law when it appears inside brackets. Expressing answers in index form keeps computations compact and comparable, exactly as scientists, journalists and government planners in	The original phenomenon asked why a small superscript number like '7' in 4.8×10^7 is so powerful for computation. This lesson demonstrates the full answer: the four laws make it possible not just to write large numbers compactly but to multiply, divide, and raise them to powers without ever expanding the full number. When Kenya's population is projected to grow by a factor of 10^2 , the answer 4.8×10^9 is reached in one step using the Product Law — exactly the kind of efficient, unambiguous

		Kenya require.	computation that makes index notation indispensable for anyone working with national-scale data.
Lesson 7 Real-Life Applications and Problem Solving — Explain & DQB Refinement	Learners observed two real Kenyan data pairs projected side by side: (1) Kenya's 2019 census population written in full (47,564,296) next to its index approximation (4.8×10^7), and (2) Kenya's 2023/24 national budget written in full (KSh 3,682,000,000,000) next to its index form ($\approx 3.7 \times 10^{12}$). Learners also observed group solution posters showing step-by-step index-law computations applied to the budget, ICT storage, and athletics contexts, and noted which laws were invoked at each step.	Learners learned that the four laws of indices (product, quotient, power of a power, zero index) are not abstract rules but practical tools used by Kenyan government planners, ICT technicians, and sports scientists to compress, compare and compute with very large and very small numbers efficiently. Specifically: the quotient law (subtract exponents) allows a journalist to calculate budget-per-person directly from standard-form figures; the product law (add exponents) allows an ICT manager to total hard-drive capacities without expanding to ordinary form; the zero-index rule ($a^0 = 1$ for any $a \neq 0$) means a scaling factor of 1 — no change — is captured elegantly as $(10^1)^0$. Learners also refined their Driving Question Board by moving remaining 'Still not sure' questions to 'What we now know' and authored personal consensus statements explaining why the tiny superscript number is so powerful.	Learners explained that scientists, economists and farmers use index (standard) form because it compresses unwieldy numbers into a compact base-and-exponent structure that follows four surprisingly neat rules — the laws of indices. These laws mean that multiplying two index expressions (same base) requires only adding two small integers, dividing requires only subtracting them, raising a power to a power requires multiplying them, and any base raised to zero always equals exactly 1. The anchoring phenomenon — the census figure 4.8×10^7 — is powerful not just because it is shorter to write, but because the exponent 7 participates directly and predictably in arithmetic through these laws, making complex computations (such as finding Kenya's budget allocation per citizen) rapid, unambiguous and easy to verify.
Lesson 8 Index Laws Map: Synthesis, Final Explanation and Portfolio Assessment	Learners revisited the two anchoring-phenomenon cards (47,564,296 vs. 4.8×10^7) displayed since Lesson 1, and conducted a gallery walk of seven Evidence Cards — one per previous lesson — showing Kenyan numerical examples of each index law, including census data, maize-yield ratios, tea-estate calculations, national-debt figures, and cell-biology measurements.	Learners consolidated all four laws of indices (multiplication: $a^m \times a^n = a^{m+n}$; division: $a^m \div a^n = a^{m-n}$; power of a power: $(a^m)^n = a^{mn}$; zero index: $a^0 = 1$) and demonstrated that each law follows necessarily from the meaning of repeated multiplication — so the superscript '7' in 4.8×10^7 is not an arbitrary shortcut but a precise instruction that participates in a consistent, powerful rule system.	Learners constructed an individual Index Laws Map synthesising all four laws with algebraic rules, repeated-multiplication reasoning, Kenyan numerical examples, and real-life Kenyan applications; completed a class written portfolio task assessing all four Specific Learning Outcomes including a written paragraph explaining the anchoring phenomenon to a novice; and wrote a self-reflection comparing their Lesson 1 prediction with their final evidence-based understanding.

ATTRIBUTION AND LICENSING

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